

Compressive behavior of thermoplastic polyurethane with an active agent foaming for 3D-printed customized comfort insoles

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ABSTRACT

The primary objective of this study was to investigate the compressive behavior of 3D-printed specimens made of a thermoplastic polyurethane filament, specifically colorFabb varioShore TPU, which incorporates a foaming agent. The foaming technology makes it possible to manipulate 3D prints' properties by adjusting process parameters that affect the level of expansion, such as printing temperature, printing speed or flow ratio. This capability opens new possibilities for 3D-printed personalized foot orthotics or for static comfort-oriented applications by tailoring the stiffness of the print based on the application requirements. Since infill density and infill pattern also affect the compressive characteristics of 3D prints, this research took these two parameters into account as independent variables, alongside the printing temperature. Thus, the specimens with printing temperatures set at 190 °C, 220 °C, and 240 °C, gyroid and honeycomb patterns, and infill variations from 10 % to 35 % in 5 % increments were experimentally investigated in compression testing. The evaluation was conducted to determine the influence of these factors on the stiffness of prints at 10 % and 20 % strains, in accordance with established standard. The findings had an important practical value as they provide data for adjusting the variable stiffness based on peak plantar pressure measurement data when developing tailored foot orthoses (insoles), which was the second objective of the research. The most influential factor affecting the compressive strength was the printing temperature, followed by infill density and pattern type. The difference in compression modulus between 10 % and 20 % strains did not exhibit statistical significance. Moreover, the main effects plots indicated minimal variations in compression modulus effects at 220 °C and 240 °C, whereas an important difference was observed at 190 °C printing temperature, which was confirmed by scanning electron microscopy investigations showing foamed layers at 220 °C and unfoamed layers at 190 °C. Furthermore, interaction plots revealed no significant interaction effect among the studied variables. Additionally, a comparison of samples' sizes and densities before and two months after testing indicated no observable modifications. When applied to custom orthotics (five pairs of insoles were customized and tested), the experimental findings documented variations in the reduction and distribution of peak plantar pressure based on the infill variability.

1. Introduction

Filaments based on active foaming technology, designed for use in Additive Manufacturing-based material extrusion process (MEX), enable the production of lightweight 3D-printed structures [1] with customizable stiffness and mechanical properties activated by manufacturing parameters settings such as printing temperature, flow ratio/extrusion

multiplier, printing speed, fan speed and layer thickness [2,3]. Foaming of elastomers can be obtained by controlled CO₂ exposure [4], by means of foaming agent [5] or by syntactic foaming [6]. This recent advancement in materials science has enabled the production of foaming filaments for MEX process, thus broadening the range of customizable properties and opening the way towards new applications in which comfort, variable hardness, stiffness and impact absorption properties

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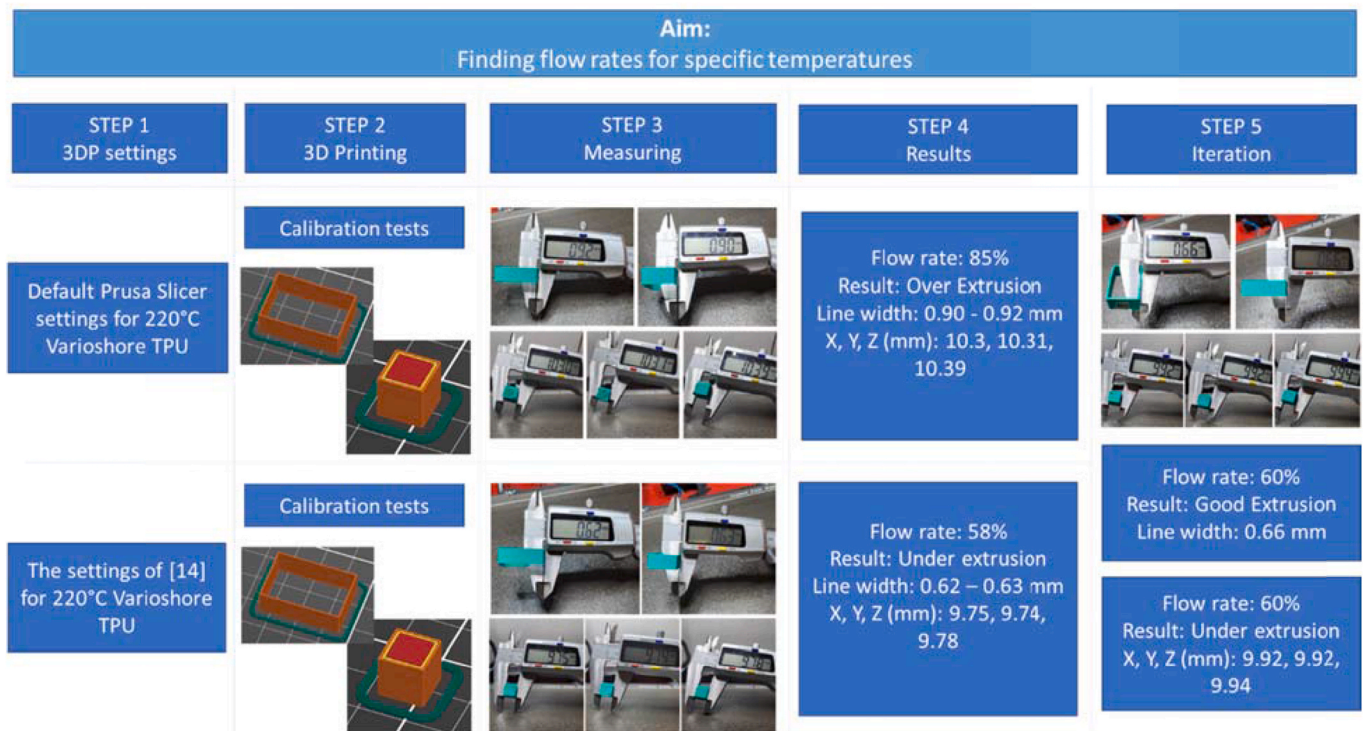


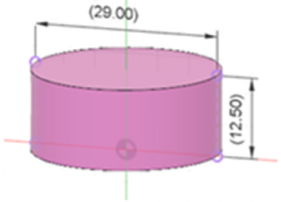
Fig. 1. General calibration method for varioShore TPU.



Fig. 2. Samples prepared for compression test.

are the main targets. These are important aspects in various industries, including footwear [7], automotive [8], packaging [9], medicine [10] or robotics [11]. However, the existing literature on this topic is still limited, with only a few studies investigating the effects of process parameters on the properties of prints produced from these novel filaments. Among the filament materials examined so far in literature, the foamed thermoplastic polylactic acid LW-PLA (Lightweight PLA) from

Table 1
Specific parameters for cylindrical specimens.

Specimen	Variable process parameters	Fixed parameters
	Printing temperature: Levels: 190 °C, 220 °C, 240 °C Infill density: Levels: 10 %, 15 %, 20 %, 25 %, 30 %, 35 % Infill pattern: Levels: gyroid, honeycomb	Bed temperature: 0 °C Flow rates: 110 %, 72 %, 60 % Layer thickness: 0.2 mm Top/bottom layers: 4 Perimeters: 0 Printing speed: 35 mm/s Fan speed: 70 % Disable for the first 4 layers, full speed at layer 6

ColorFabb [1,3,12] received more attention than thermoplastic polyurethane (TPU) with foaming agents such [13–16].

In comparison to insoles' conventional production techniques and materials, MEX provides an additional customization level to the user's matching foot shape and size, providing relevant features such as adjustable support, cushioning, and tailored stiffness for pressure distribution based on individual foot conditions and activity needs, harder or impossible to achieve otherwise. The use of MEX process for producing insoles was reported in studies such as [17–19] for tailoring design and stiffness based on the user's foot characteristics and anatomy, weight and use purpose (medical, comfort). However, implementing large-scale production of customized 3D-printed insoles poses several challenges. These include the difficulty of 3D printing highly flexible TPU filaments, which necessitates careful calibration of the 3D printer and precise adjustment of flow ratio and printing temperature settings [20]. Additionally, there is the aspect of the availability of equipment and dedicated software for peak plantar pressure



Fig. 3. Compression testing of a varioShore TPU specimen.

measurement, 3D scanners and software for tailoring the design and 3D modeling the insoles. Furthermore, there is a lack of comprehensive data on the compression properties of various TPU filaments recommended by their producers for insole production, as well as very little information on the correlation between 3D printing process parameters, compression strength, and plantar pressure. This study aims to tackle these issues by proposing a low-cost approach for customizing insole design and static peak plantar pressure measurement. All these in addition to answering the following research question that constitutes the primary objective of the paper: How do different configurations of printing parameters affect the compressive strength of 3D prints produced from varioShore TPU using MEX process? The acquired experimental data were subsequently used to fabricate insoles with adjustable stiffness based on the peak values of plantar pressure measured with dedicated equipment based on the proposed work approach.

Technical data provided by varioShore TPU producers (ColorFabb and Lubrizol Engineered Polymers, NL) indicates that this material's hardness can be adjusted between shore 55A (fully foamed) and shore 92A (unfoamed, before 3D printing) by varying the nozzle (printing) temperature within the range of 190–250 °C, and it can expand to approximately 1.4–1.6 times its original volume [21]. These properties can be achieved within a single print (by G-code modifications), allowing for a wider range of applications. However, identifying the appropriate parameter settings for the desired outcomes is a complex task due to only a few studies correlating printing temperature, infill density, speed, and flow rate with properties such as hardness and compressive strength. In this context, this paper aimed to address this knowledge gap by providing insights into the relationship between three process parameters and the resulting varioShore TPU specimens' compressive properties. These parameters are printing temperature with three levels (190 °C, 220 °C, 240 °C), infill density with five levels (10 %, 15 %, 20 %, 25 %, 30 %, 35 %) and two infill patterns (hexagonal/honeycomb, gyroid).

In [7], calibration tests were conducted for studying varioShore specimens using increments of five units for nozzle temperature, flow

rate, printing speed and fan speed. The main conclusion was that printing temperature and printing speed have the most significant influence on the dimensions of the wall of the calibration cube. In our investigation, the selection of the specific levels, as well as the calibration of process parameters prior to specimen production were based on a limited number of previous investigations in the field. In Ref. [13], was examined how the printing temperature affects the modulus of elasticity and tensile strength of specimens with 100 % infill density, as well as the absorption capacity and peak stress of honeycomb lattice structures. In Ref. [20], the effect of printing temperature on specimens' hardness was investigated. The main findings confirmed the conclusion drawn in Ref. [14] that a temperature of 220 °C results in maximum foaming. In Ref. [15], varioShore was characterized from several perspectives such as static compression strength for different insoles' thicknesses, coefficient of friction and specific gravity.

Better comfort was reported in Ref. [22] for varioShore insoles 3D-printed at 220 °C than at other temperatures (190 °C, 240 °C), as well as insoles resistance to over 300,000 flexural cycles. While acknowledging the existence of other commercial such as Filaflex Foamy from Recreus [23] or Nanovia Flex EXP [24] that includes expanding microcapsules within a nylon matrix – recommended by their producers for footwear/insoles production, and non-commercial expandable filaments for MEX process [16], the focus of the current study is varioShore filament as it has previously been shown to withstand a high number of flexural fatigue cycles [20], this being an important functional criterion for using such filament in insoles production. In this sense, the current research represents a step forward in advancing of the understanding the properties of this material and for optimizing the 3D printing process parameters for its application in orthotics.

Summarizing, this research proposed several novelties, starting with the investigation of the compressive characteristics of MEX specimens produced from varioShore TPU, the printing temperature being used to activate the foaming agent. The impact of infill-related parameters and printing temperature on compression behavior was analyzed. The findings were used to produce insoles with variable stiffness corresponding to various plantar pressure values. A methodology to design insoles, measure plantar pressure, and set printing parameters based on acquired data was also developed and applied to a healthy user who tested five pairs of insoles with different stiffness zones. Both the static plantar pressure distribution concerning the barefoot and the subjective comfort levels associated with the conditions under examination were assessed.

2. Materials and method

2.1. Calibration process

In this research, the filament of choice was green varioShore TPU with a diameter of 1.75 mm. A commercial filament was used because it is very important, in our view, to ensure that podiatrists have access to cost-effective materials and technology (including 3D printers, scanners, and modeling software) to provide a wider range of users with access to products that can enhance their comfort.

Prusa MK3s+ with a direct drive E3D Revo extruder was used for manufacturing the specimens for compression testing and insoles. Before 3D printing, a calibration procedure was conducted on the extruder which firstly included adjusting the tension of the feed gears as detailed in a prior study [20]. This step was mandatory due to the flexible nature of the specified filament, causing quality problems of the prints, filament jams, clogging etc.

The calibration had as main purpose setting a correlation between the flow rate value and the printing temperature. The input data for varioShore TPU printing settings were gathered from the default Prusa Slicer profile. As shown in Fig. 1, illustrating the general framework of the calibration process, the flow rate for the 220 °C nozzle temperature was established at 85 % from the Prusa settings and 58 % from the

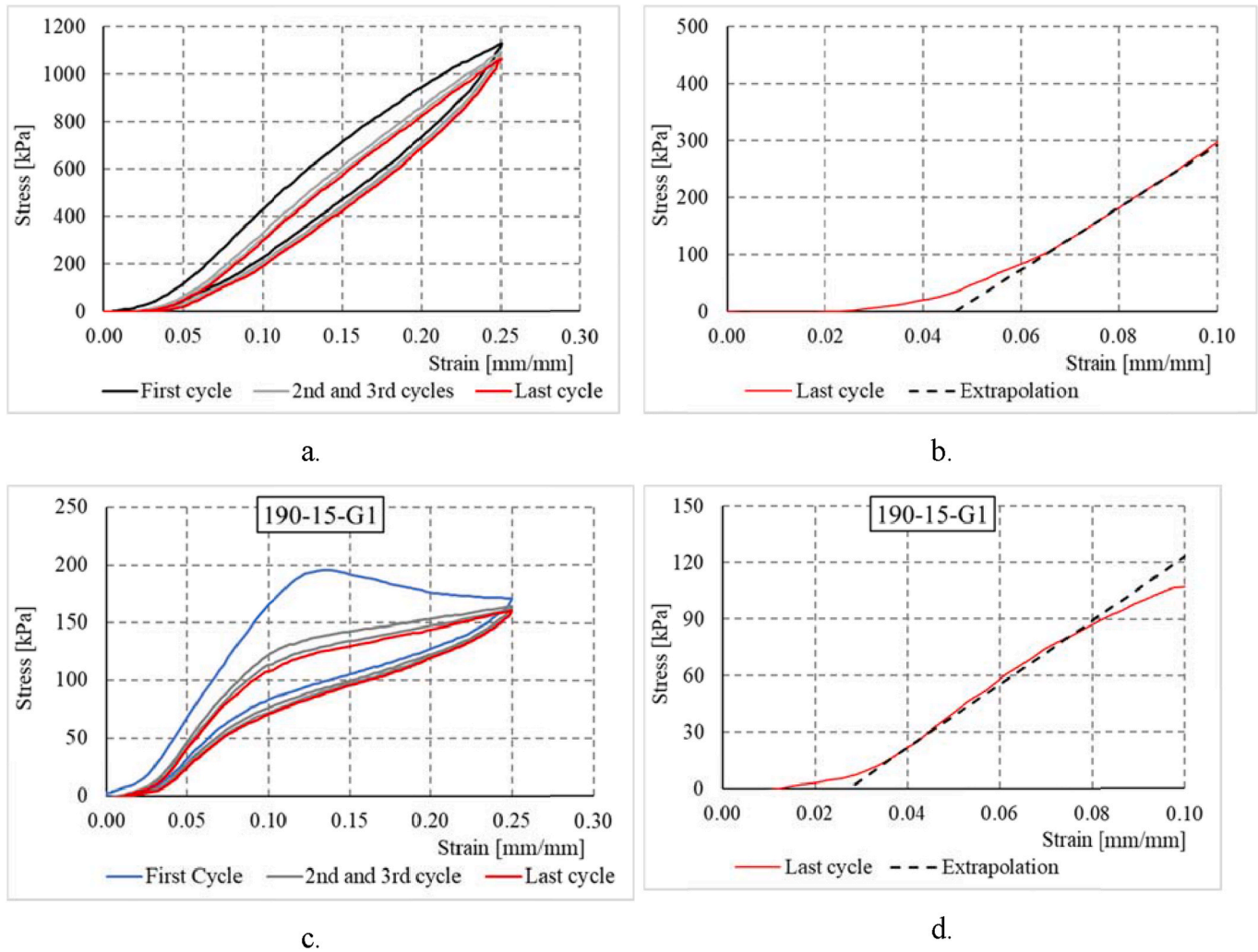


Fig. 4. Example of derived stress-strain curve, gyroid sample 3D-printed at 190 °C, 35 % infill; b) Extrapolation of the real start of the fourth cycle; c) Example of stress-strain curve, gyroid sample printed at 190 °C, 15 % infill; d) Extrapolation of the true start of the fourth cycle.

calibration performed in Ref. [14].

The initial calibration was conducted on calibration cubes through a process of incremental adjustment of the flow rate. However, even though the flow rate was initially configured at 125 % for 10 mm × 10 mm × 10 mm cubes at 190 °C to ensure accurate dimensions, the first test with a larger object (i.e. an insole), revealed that this flow rate caused undue pressure within the nozzle, leading to filament jams. As a result, the flow rate had to be recalibrated to 110 % to mitigate this issue. This observation emphasizes the difficulties in determining the ideal value for flexible filaments, which can also vary based on the size of the printed component.

2.2. 3D printing specimens

The compressive tests were conducted on cylindrical specimens with 29 mm diameter and 12.5 mm height. Compression-strain characteristics were determined according to ISO 7743:2017 standard specifications for compression testing of polymeric materials [25] as in León-Calero et al. [26] who tested different types of TPU specimens with grid, gyroid and honeycomb patterns and infill densities 0%, 20%, 50%, 80% and 100%.

Fig. 2 shows the samples prepared for compression testing. The difference in color is due to the different printing temperature which modifies the foaming/internal porosity of the filament.

Table 1 displays the process parameters investigated in this study

and their respective levels, with three specimens for each configuration. 108 samples were tested in total.

3D printing fixed process parameters in Table 1 were set based on the calibration step described in section 2.1, which provided the optimal flow rate values for each printing temperature, as well as on previous experience with this material [20,22]. The flow rate value varied with different printing temperatures, but it remained constant for each specific temperature setting.

2.3. Compression testing

Uniaxial compression tests were conducted along z axis of the cylindrical specimens (three for each sample group, with the process parameters presented in Table 1) on Lloyd Instruments LRX Plus (Ametek, Inc., UK) with a load cell of 5 kN. The upper and lower surfaces of the samples were lubricated to reduce friction between the sample and the installation fixtures.

A speed of 10 mm/min was applied for all samples until reaching 25 % of strain according to standard, at which point the strain was then released at the same rate (Fig. 3), as recommended in ISO 7743:2017. The procedure of compression and release was performed three additional times, resulting in a total of four uninterrupted compression cycles, recording being made as force-deformation curves. These were converted in stress-strain coordinates by dividing the force to the initial cross-section and the deformation to the initial length.

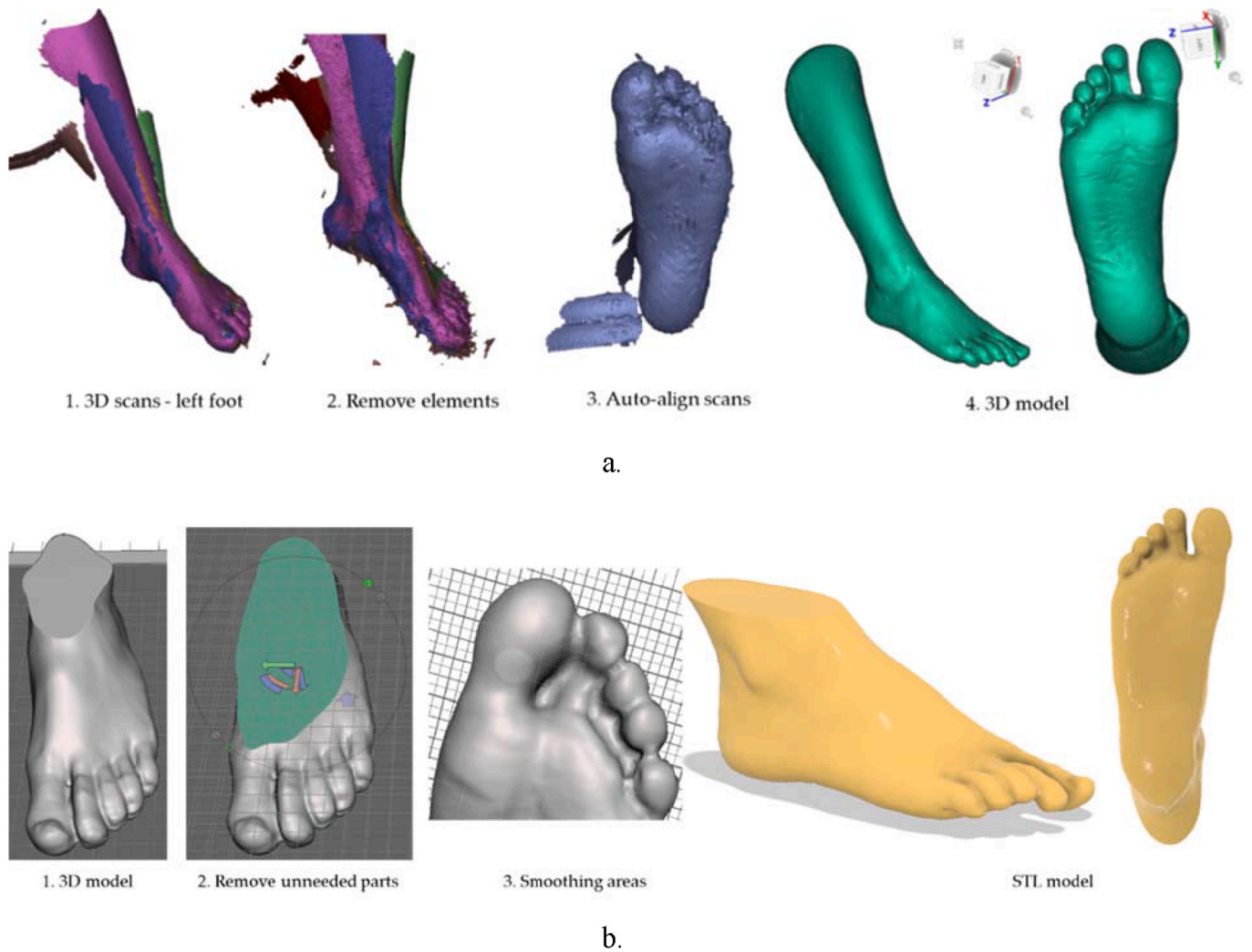


Fig. 5. 3D scanning with Artec Eva 3D scanner (a); reconstructed model of the foot (b).

A typical stress-strain curve from the results is presented in Fig. 4a.

All load cycles presented hysteresis loops, and an incomplete recovery at the end of each cycle, suggesting viscoelastic behavior. The secant compression modulus was extracted for the last loading cycle. Due to the forming of a “toe region” at the beginning of the cycle, the real start of the fourth one was determined by extrapolation of the load curve in the $0.1 \cdot \sigma_{\max}$ and $0.2 \cdot \sigma_{\max}$ (Fig. 4b) which was determined to be linear.

Consequently, the moduli were extracted at 10 % and 20 % strain from the extrapolated starting point, following relation (1).

$$E_i = \frac{\sigma_i}{\varepsilon_i} \quad (1)$$

where.

E_i – Compression modulus at 10 % and 20 % strain, expressed in [MPa]

ε_i – Strain formed during the last load cycle, expressed in [mm/mm]

σ_i – Recorded stress at which ε_i is produced, expressed in [MPa]

2.4. Customizing the insoles design and their 3D printing process

The process of designing the insole involved 3D scanning the patient's foot (Fig. 5a is an example for the right foot). This method of data acquisition ensured a customized fit tailored to the unique contours of

the individual's foot. Following this, the reconstructed foot model was used as a negative mold to shape the design of the insole. This specific modeling process was accomplished either through Gensole web-based software (Gyrobot, UK) dedicated to insole design and production through 3D printing.

The 3D scanning was carried out with Artec Eva Lite handheld 3D scanner (Artec, USA), and the acquired data was processed using Artec Studio 17 Professional. Various processing options were applied to the points cloud, including tracking geometry to facilitate scanning, model creation for adjusting scan quality, the editor for eliminating undesired elements, auto-align scans for optimal positioning, and exporting to mesh to create an STL model of the foot. To enhance the model's quality, further processing was performed in Meshmixer (Fig. 5b).

Fig. 6 presents the workflow in Gensole which produced the digital 3D models of the insoles.

The steps in Figs. 5–6 are part of the customization workflow not only for ensuring fit to user's foot, but also for delineating the pressure zones which subsequently will be used for setting in slicer the zones of various infill densities and patterns.

2.5. Peak plantar pressure measurements

For peak plantar pressure measurement, an electronic insole with 16 pressure sensors covered in transparent silicone was used (Fig. 7a). According to the manufacturer's specifications [27], the pressure sensors become active when subjected to a weight of minimum 500g, and

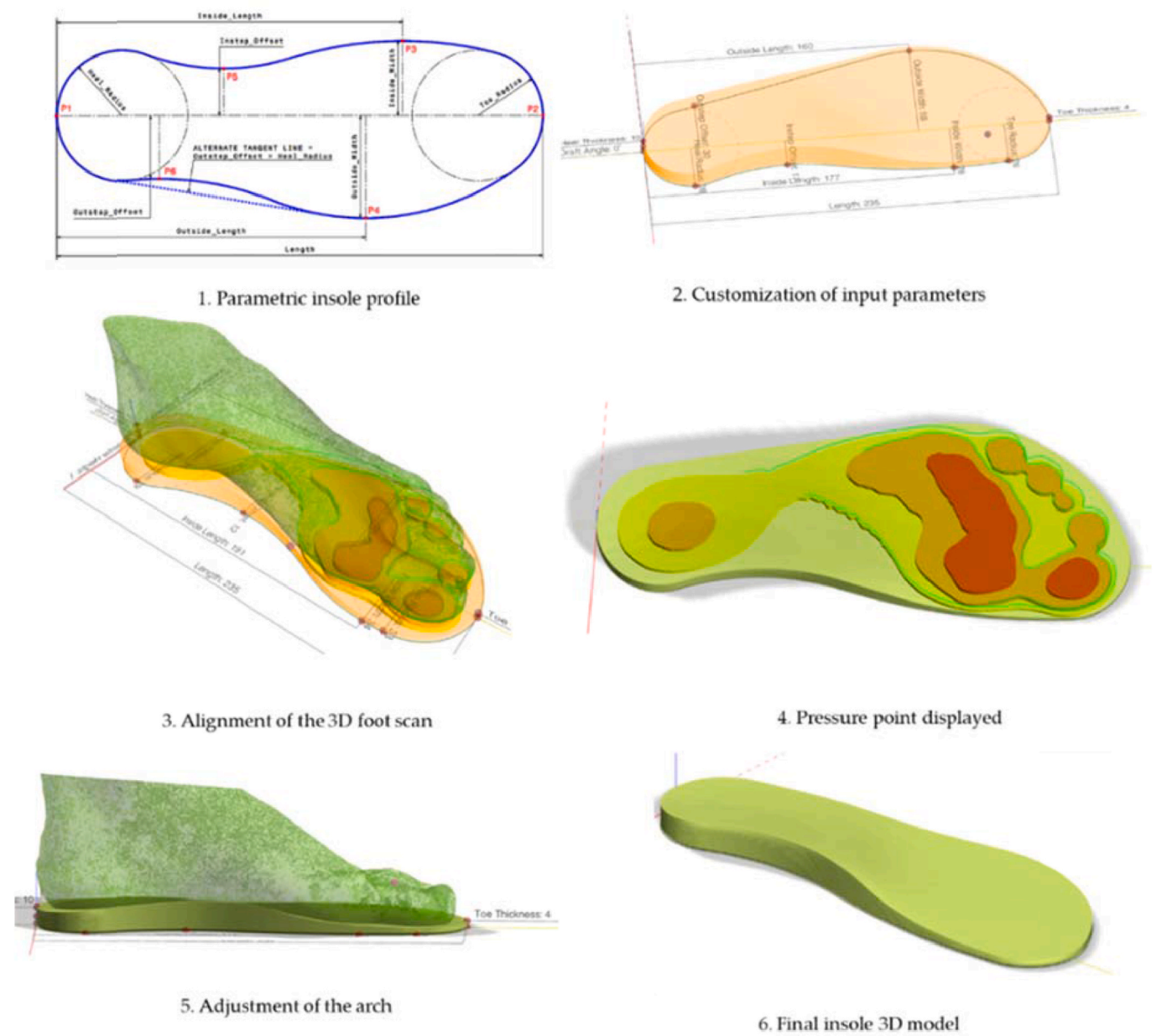


Fig. 6. Customized insole design workflow using Gensole.

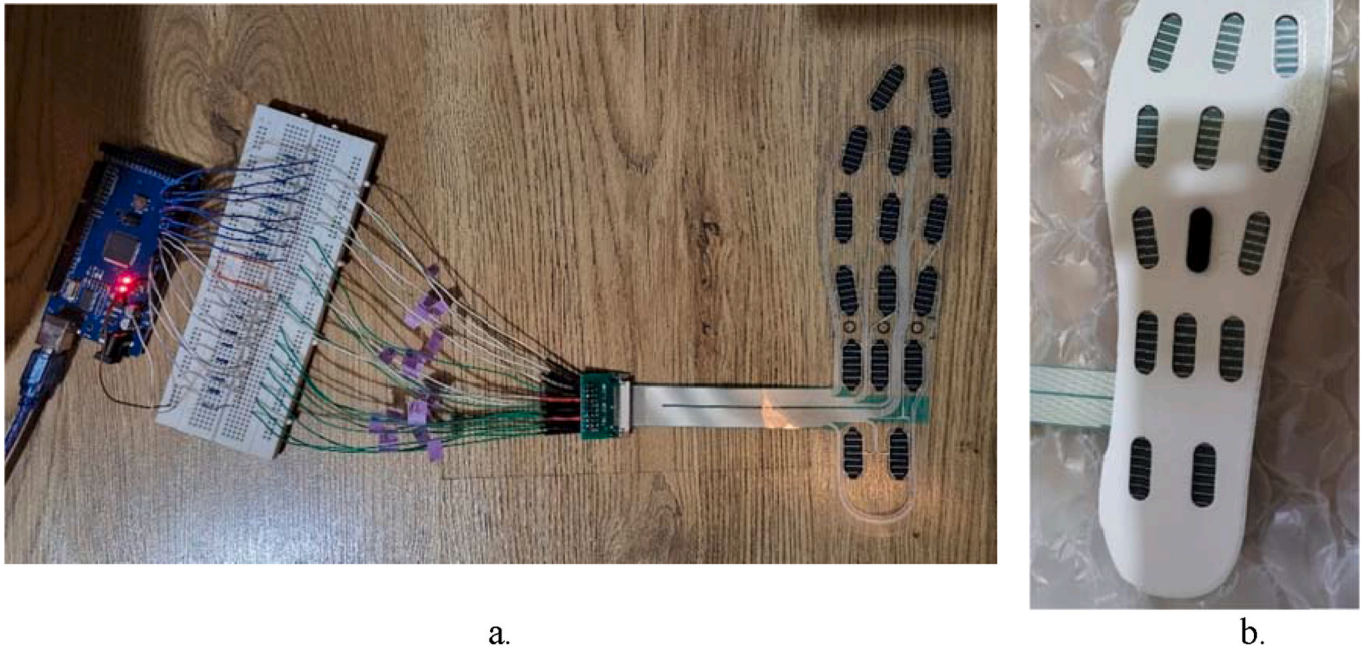


Fig. 7. Sensor insoles: a. Developed peak plantar measurement system, b. PETG frame for data reading.

Table 2
Mean values and standard deviations of compression moduli for the tested samples.

Specimen	Compression modulus [MPa] – mean at 10 % strain, G	Compression modulus [MPa] – mean at 20 % strain, G	Compression modulus [MPa] – mean at 10 % strain, H	Compression modulus [MPa] – mean at 20 % strain, H
190–35	5.17 ± 0.25	4.88 ± 0.25	8.56 ± 0.34	9.08 ± 0.35
220–35	1.68 ± 0.06	1.59 ± 0.05	2.6 ± 0.10	3.02 ± 0.10
240–35	1.82 ± 0.03	1.75 ± 0.03	2.94 ± 0.13	3.19 ± 0.13
190–30	4.35 ± 0.15	3.91 ± 0.15	8.18 ± 0.37	8.54 ± 0.35
220–30	1.45 ± 0.15	1.33 ± 0.15	2.38 ± 0.37	2.76 ± 0.35
240–30	1.47 ± 0.09	1.41 ± 0.08	2.82 ± 0.17	3 ± 0.20
190–25	3.2 ± 0.12	2.57 ± 0.13	6.87 ± 0.31	7.14 ± 0.30
220–25	1.13 ± 0.09	0.97 ± 0.07	1.96 ± 0.15	2.26 ± 0.16
240–25	1.19 ± 0.05	1.11 ± 0.04	2.25 ± 0.29	2.42 ± 0.27
190–20	2.19 ± 0.07	1.46 ± 0.07	5.95 ± 0.31	5.79 ± 0.33
220–20	0.77 ± 0.04	0.59 ± 0.03	1.62 ± 0.17	1.79 ± 0.21
240–20	0.84 ± 0.03	0.75 ± 0.03	1.7 ± 0.27	1.84 ± 0.23
190–15	1.19 ± 0.03	0.71 ± 0.04	4.06 ± 0.28	3.26 ± 0.23
220–15	0.43 ± 0.03	0.31 ± 0.02	1.47 ± 0.27	1.59 ± 0.31
240–15	0.51 ± 0.02	0.4 ± 0.02	1.36 ± 0.19	1.46 ± 0.19
190–10	0.42 ± 0.03	0.27 ± 0.03	1.65 ± 0.07	1.28 ± 0.05
220–10	0.2 ± 0.02	0.14 ± 0.01	0.64 ± 0.06	0.54 ± 0.05
240–10	0.18 ± 0.01	0.14 ± 0.01	0.74 ± 0.10	0.69 ± 0.11

are able of tracking up to 10 kg.

An Arduino Mega 2560 with an ATmega 2560 microcontroller was used as hardware, featuring 16 analog inputs to connect all pressure sensors. Also, a Flexible Printed Circuit (FPC) – 20 Pins with 1.0 mm pitch, wires, bread board and 470 kΩ resistors was used as standalone development board. To ensure stability and consistent positioning of the electronic insole, a frame was designed, and 3D printed from PETG (Fig. 7b). This template does not contact the electric insole – there is a 0.2 mm gap between parts to avoid interfering with the measurement.

The sensors were pressed using a 3D printed pad made of TPU

Filaflex 70A with a density of 100 %. Programming was carried out using Visual Studio Code with the Python extension and the Arduino IDE. Sensors calibration was conducting with masses of 500g, 1500g, 3000g and 4500g, subsequently transformed into plantar pressure values. The data obtained from these calibrations were then refined and adjusted using an algebraic, polynomial function. The sensors' data were converted into corresponding weights, which were subsequently translated into plantar pressure results. This data helped identify zones with peak plantar pressures, guiding adjustments in infill density and pattern to redistribute pressure and enhance overall insole comfort.

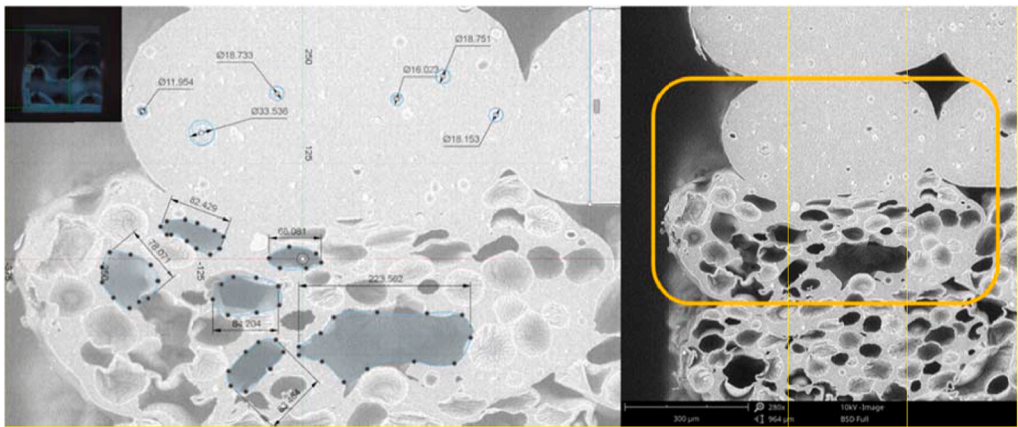
A limitation of our present study is that the peak plantar measurement system developed, while cost-effective and significantly less expensive than dedicated equipment (such as the in-Pedar system in Ref. [22]), is currently unable to assess plantar pressure during dynamic activities like walking or running. Only static pressure data was collected. As a result, the developed insoles are recommended primarily for enhancing comfort during prolonged standing rather than dynamic movements. Ensuring comfort for individuals working long hours in a static standing position is emphasized in various studies [28,29] as very important for preventing health issues.

3. Results and discussions

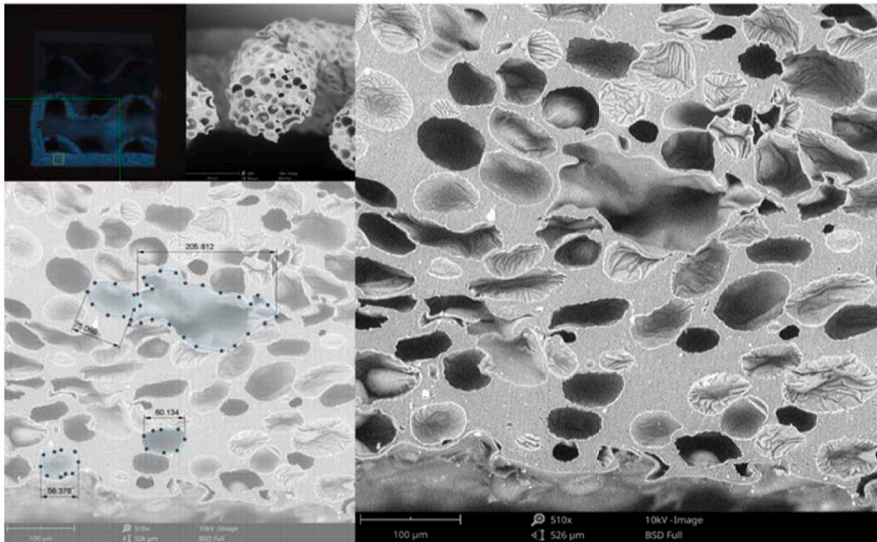
3.1. Compression test results

Table 2 is a synthesis of the experimental data gathered through compressive testing. Specimens are denoted as follows: temperature-infill, meaning that 190–35 indicates a specimens 3D printed at 190 °C with 35 % infill density, while the letter (G or H) indicated gyroid or honeycomb pattern.

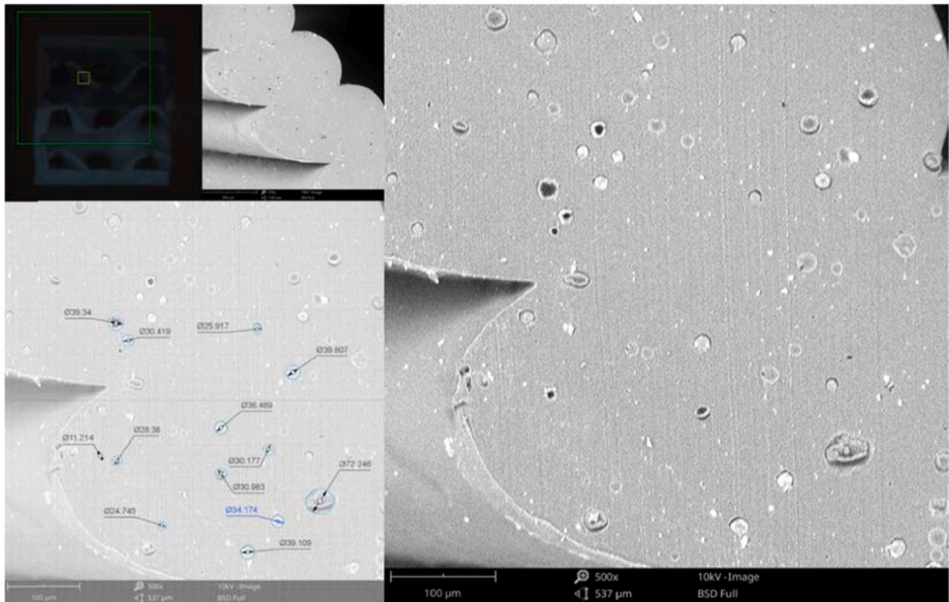
As shown in Fig. 4, varioShore, as other TPU materials, displays significant strain viscoelastic response, characterized by hysteresis in their stress-strain behavior. The examination of the compression process of the varioShore samples indicated that buckling and bending, rather than cracking or permanent deformations were the primary mechanisms



a.



b.



c.

Fig. 8. SEM investigations: (a) sample with layers 3D-printed at 190 °C and 220 °C; (b) voids in a 220 °C filament thread caused by the expansion of the foaming agent; (c) internal structure for 190 °C.

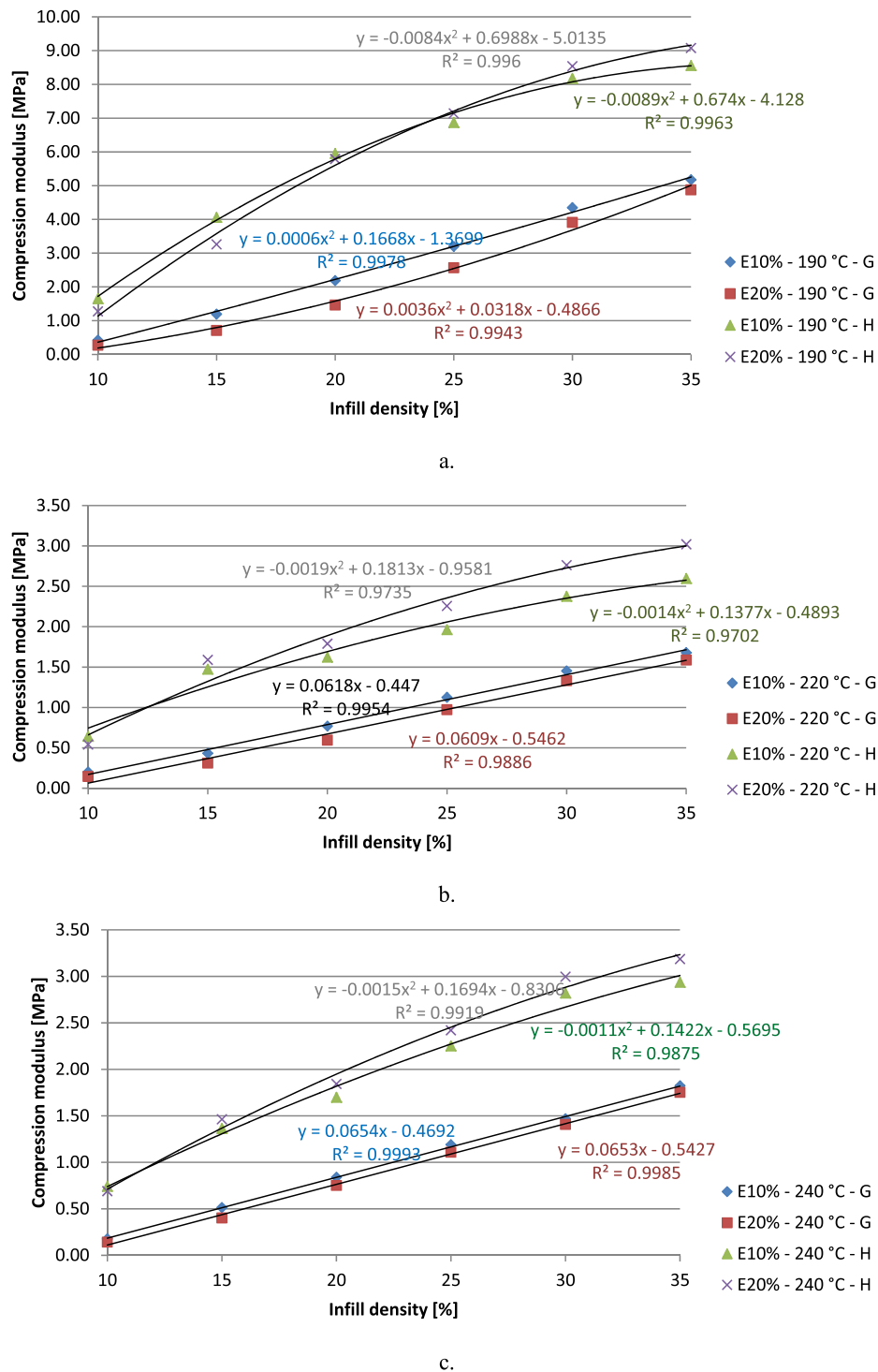


Fig. 9. Compression moduli for different process parameter configurations: a. 190 °C samples; b. 220 °C samples; c. 240 °C samples.

responsible for the structures' deformation, as observed also in Ref. [30].

It can be noted that the compression strength is the highest for 190 °C samples, followed by those at 240 °C and 220 °C. The difference in compressive strength between the samples printed at 220 °C and 240 °C is considerably smaller compared to that between samples printed at 190 °C and 240 °C. This indicates that the highest foaminess is achieved at 220 °C, whereas samples printed at 190 °C remain mainly unfoamed. This observation is supported not only by the specifications provided by the filament producer but also by findings from a previous study [20]

where it was found that the shore A hardness of samples printed at 190 °C is the highest, followed by samples 3D-printed at 240 °C and 220 °C. Additionally, Scanning Electron Microscopy (SEM) investigations conducted using Phenom ProX (Phenom-World B.V., NL) microscope revealed the unexpanded state of samples 3D-printed at 190 °C and the presence of larger pores in samples 3D-printed at 220 °C. Layers 3D-printed at 190 °C and layers 3D-printed at 220 °C were deposited within the same sample for SEM analysis. Fig. 8a highlights the difference in the layers structures: in the 190 °C layers, the foaming agents are characterized by well-defined pores, whereas in the 220 °C

Table 3

Samples' mean density before testing and 2 months after testing, and 3D printing time.

Specimen	Mean density (kg/cm ³)				3D printing time (min)	
	G		H		G	H
	Before test	2 months after	Before test	2 months after		
190–35	0.532	0.532	0.620	0.621	49	57
220–35	0.295	0.300	0.339	0.345	49	57
240–35	0.330	0.331	0.377	0.381	49	57
190–30	0.484	0.487	0.568	0.569	44	53
220–30	0.266	0.272	0.317	0.321	44	53
240–30	0.307	0.307	0.359	0.362	44	53
190–25	0.437	0.438	0.510	0.510	41	47
220–25	0.255	0.246	0.281	0.284	41	47
240–25	0.273	0.275	0.321	0.321	41	47
190–20	0.390	0.391	0.453	0.453	36	42
220–20	0.215	0.219	0.240	0.242	36	42
240–20	0.246	0.248	0.289	0.292	36	42
190–15	0.342	0.344	0.401	0.402	31	37
220–15	0.186	0.188	0.230	0.232	31	37
240–15	0.214	0.217	0.247	0.250	31	37
190–10	0.303	0.304	0.326	0.327	27	29
220–10	0.165	0.166	0.179	0.181	27	29
240–10	0.193	0.193	0.205	0.206	27	29

Table 4

ANOVA analyses.

Factors	P-value
10 % compression modulus at for gyroid and 20 % compression modulus for gyroid	0.632
10 % compression modulus at for honeycomb and 20 % compression modulus for honeycomb	0.901
10 % compression modulus at for gyroid and 10 % compression modulus for honeycomb	0.0199
20 % compression modulus at for gyroid and 20 % compression modulus for honeycomb	0.0067
3D printing times for gyroid and honeycomb samples	0.043
Means density of gyroid vs. honeycomb samples	0.232
Mean density before and 2 months after testing for gyroid samples	0.968
Mean density before and 2 months after testing for gyroid honeycomb	0.960

layers, they are expanded and substituted with almost spherical voids (Fig. 8b). At 190 °C, the largest size of the expanded pores is ~33.5 µm, and at 220 °C it can even reach ~223.5 µm by joining two or more pores. For another area (Fig. 8c), at 190 °C the maximum pore diameter is ~72.5 µm and at 220 °C for the area in Fig. 8b the maximum size is ~206 µm.

Honeycomb pattern showed superior compression performance compared to the gyroid pattern regardless of the printing temperature (Fig. 9), while both patterns required nearly identical printing times (e. g., 29 min - 190 - 10 % - honeycomb; 27 min - 190 - 10 % - gyroid, Table 3). As mentioned in Introduction, studies on compressive properties of TPU with foaming agents for MEX process are lacking. Therefore, the findings in this research cannot be directly compared with those from similar investigations. However, broadening the range of material from foamed TPU to TPU for MEX, previous studies confirmed that gyroid has lower compressive strength than other patterns such as cubic, triangle or rectilinear, as shown in Orsu et al. [19] who explored the compressive strength of PolyFlex TPU specimens with varied infill patterns and densities with the purpose of customizing different insole zones based on experimental outcomes. Omer et al. (2022) studied the mechanical properties of ultra-flexible TPU samples (60A shore) for different printing temperatures, revealing peak tensile strength and

elongation at break at 230 °C, and maximum storage modulus at 220 °C. In Ref. [31] quasi-static and cyclic compression tests on graded honeycomb custom-designed structure from NinjaFlex TPU were conducted, showing that the material softens after 50 cycles and the damping capacity can be thus reduced. In Ref. [22], varioShore insoles proved a very good resistance to a large number of flexion cycles, with no infill structure failures and no remnant deformations in the zones of fixation in the testing equipment. Regarding the influence of process parameters such as layer thickness and printing temperature on specimen mechanical performance at different strain rates, the study of Vidakis et al. [32] can be mentioned. Their results emphasized the significant influence of printing temperature, particularly notable for TPU filaments with foaming agents activated by temperature. In Ref. [33], the impact of compression strain rate over TPU specimens with 100 % infill density showed that no cracks were developed not even at the highest strain of 200 mm/min, proving material suitability for applications in which mechanical durability is required.

Compression test results were plotted in Fig. 9. This data can be used to predict the value of compression modulus at different infill densities and printing temperature, with coefficients of correlation exceeding 0.97. There is an almost linear relationship between the infill density and compression modulus across all cases, with polynomial functions demonstrating very low x^2 coefficients. Similar findings regarding the linear dependency between compressive strength and infill density (at levels of 50 %, 75 %, and 100 %) were also reported by Dixit et al. [34] for TPU materials.

The largest compression modulus was 9.08 MPa for 190-35H specimens, while the smallest values were obtained for 10 % and 15 % infills density and temperatures of 220 °C and 240 °C. When the infill is below 15 % and the printing temperature over 220 °C, the infill pattern type is not relevant (the interaction plots presented further sustain this observation).

Table 3 provides data on the density of the samples, indicating that honeycomb samples exhibit higher density. Additionally, it shows that the printing temperature impacts the level of foaming within each sample, which influences the compression strength, as also proved by SEM. To characterize the behavior of the material in time, the samples density was measured 2 months after the compression testing, and differences were analyzed.

One-way ANOVA with significance level of 0.05 was used for a better view over the parameters influence over compression moduli (Table 4). It could be noted that the compression modulus values at 10 % and 20 % strains were similar for similar pattern types, but they differ between patterns. In Table 4 are also presented ANOVA results for the 3D printing time for the samples with gyroid vs. honeycomb infill patterns, and for assessing if there is a significant difference between the samples' density before testing and after 2 months of shelf-life. The results showed that there is a significant difference between the densities of samples with gyroid vs. honeycomb infill pattern. At the same time, there is no significant difference between samples densities before and 2 months after testing which means that the samples structure was not affected during testing (samples dimensions remained the same as it can be seen in the supplementary material).

The main effects plots displayed in Fig. 10 (Minitab, UK) showed the impact of the examined independent variables on the compression modulus, represented by the connecting lines between the parameter levels. It can be noted that the printing temperature has the highest relevance. The horizontal or nearly horizontal lines indicated the absence of a significant main effect (i.e. the effect of the parameter is not statistically significant). There are only slight variations in compression modulus effects observed at 220 °C and 240 °C, while a considerable difference was noticeable when the printing temperature is set to 190 °C.

Pareto charts in Fig. 11 also showed that the compression modulus is most significantly influenced by the printing temperature, followed by infill density and infill pattern. For the 20 % strain values, the difference in the influence of variables is not as evident as for the 10 % strain. This

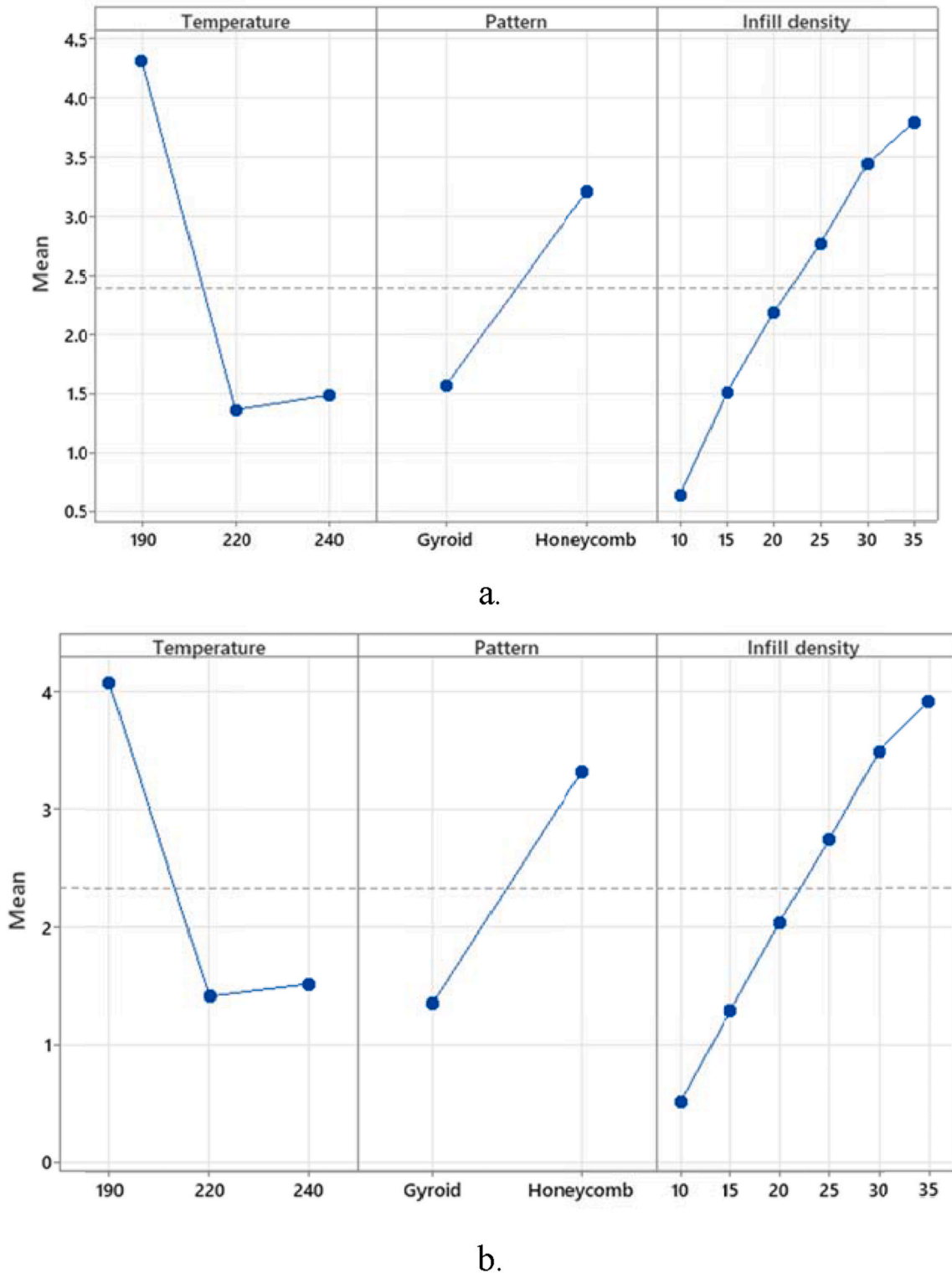


Fig. 10. Main effects plots for 10 % strain (a) and 20 % strain (b) for compression modulus.

shows that the more the samples are compressed, the less important their infill structure is due to deformation.

Infill pattern was reported as influencing the compressive strength of TPU samples in Ref. [8] for gyroid and Cross3D, and being less relevant for concentric and Cross2D. The study also analyzed infill densities up to 30 % for understanding the impact of this parameter over the stiffness in comparison with viscoelastic polyurethane foams, traditionally used for pressure relief/comfort applications such as car seats, wheelchair seats,

bed mattresses.

Fig. 12 represents the interaction plots which are a visual illustration on how the analyzed factors combined effects reflect on the compression modulus of varioShore TPU specimens. It can be noted that there is no interaction effect among parameters (the lines are parallel), moreover for the specimens 3D printed at 220 °C and 240 °C, the effects overlap.

From prior investigations [22], the evaluation of insoles 3D-printed from varioShore with similar infill density, but 3D-printed at 220 °C and

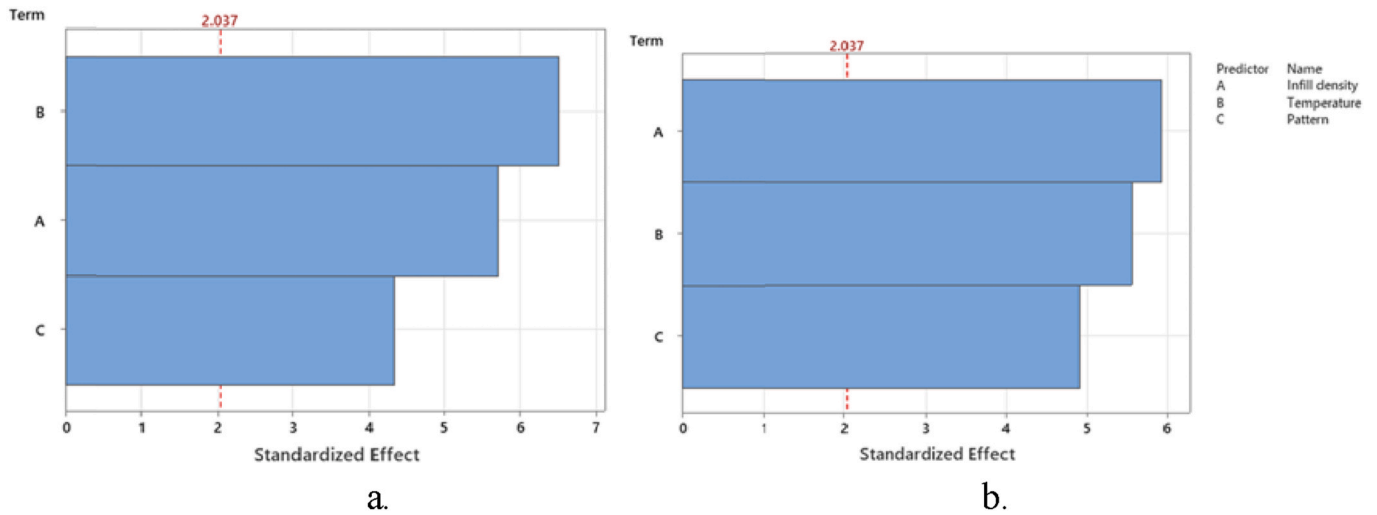


Fig. 11. Pareto charts of parameter effects on the compression modulus of varioShore specimens.

240 °C revealed better comfort provided by the insoles produced at 220 °C, as reported by the user. As a result, in Fig. 13 are shown the stress-strain curves for various densities at 220 °C: the gyroid infill pattern (Fig. 13a) and the honeycomb infill pattern (Fig. 13b). These were further to set variable infill densities in different zones of the foot. To this end, correlations between the values of the peak plantar pressure (section 3.2) and infill densities were established with the purpose of reducing and uniformizing the static peak plantar pressure.

3.2. 3D-printed insoles testing results

In previous sections, a procedural framework for podiatrists was presented. A user weighing 48 kg was used to illustrate how the information from mechanical tests can be applied. A similar methodology can be developed for other individuals with different weights and plantar pressure characteristics by following the provided workflow.

Using the data acquired from the experimental trials with the sensors insole, a bare foot measurement of plantar pressure was firstly conducted (Fig. 14).

Three configurations of infill density-infill pattern (V1–V3) were selected based on the experimentally determined compression values, corresponding to three targeted peak plantar pressure values: 161.84 kPa, 91.67 kPa and 53.47 kPa (Fig. 14). Additionally, two more insoles with uniform infill density, medium stiffness, were 3D-printed to be used as references: 20 % gyroid infill (V4), 20 % hexagonal infill (V5).

The 3D printing times for V1–V5 insoles (for each foot) were the following: V1 - 3h 15 min, V2 - 3h 13 min, V3 - 2h 58 min, V4 - 3h 13 min, V5 - 2h 43 min.

In this study focused on developing static comfort insoles, three displacement ranges were selected: 0.06 mm, 0.08 mm, and 0.2 mm. These values represent the extent to which the heights of the insoles are reduced under the user's weight. However, here it should be noted that the podiatrist should establish the values based on a throughout analysis of the patient pathology and individual characteristics.

Fig. 15 was used for selecting the infill pattern – infill density combination corresponding to different plantar pressure zones (Table 5). The pattern and the density were selected around the zones where the horizontal lines of plantar pressure intersect the stress-strain curves and the vertical lines corresponding to different displacement values.

Fig. 16 captures different phases of variable infill insoles manufacturing for illustrating the structure within different zones of the insoles, also highlighting the zones with diverse infill configurations. All insoles were 3D printed at 220 °C as this printing temperature provides the best comfort as reported by the user in Ref. [20].

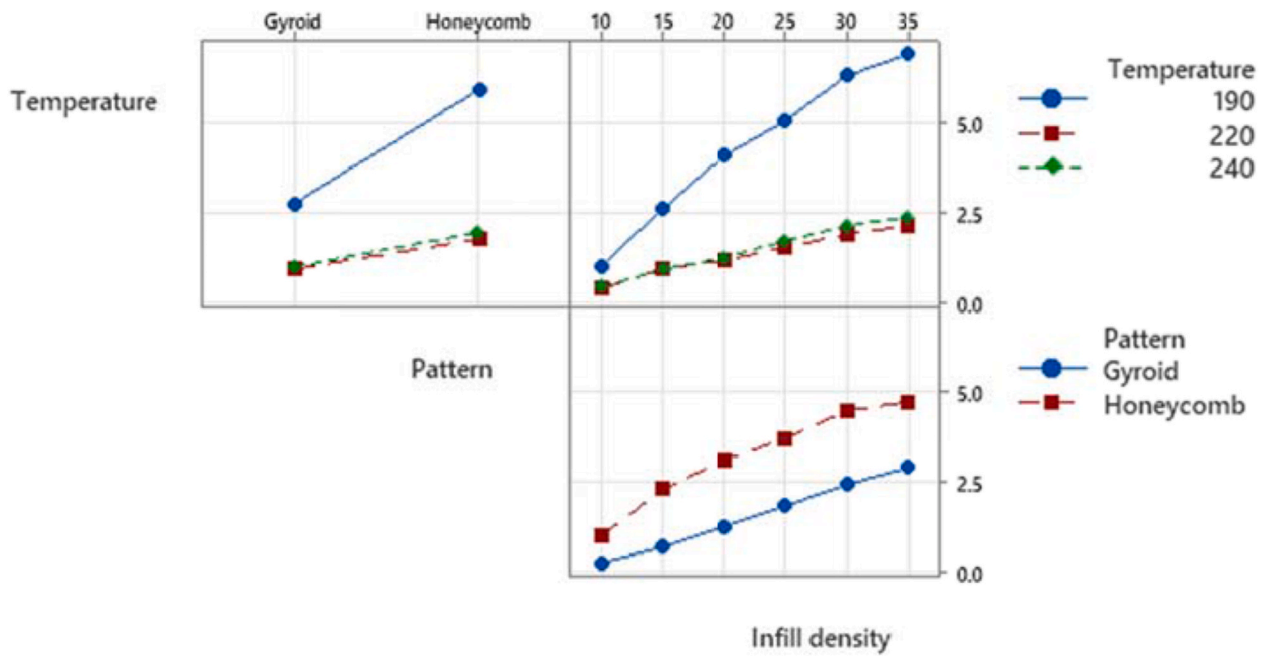
The 3D-printed insoles underwent testing using the electronic insole setup to evaluate their effectiveness in reducing peak plantar pressures. Tests showed that the insoles resulted in a significant reduction in plantar pressure (Fig. 17).

The results were summarized in Table 6 for comparison with the peak plantar pressure of barefoot in two important zones: hindfoot (heel) and forefoot. Conducting ANOVA for values in Table 5 (p-value = 0.677 for V1–V5, p-value = 0.876 for V1–V3 insoles with variable infill), it was found that there is a significant difference between the peak plantar pressure values. This shows the importance of thoroughly selecting the infill parameters. In this sense, further research will consider the development of a finite element-based procedure for insoles, to support the selection of the pattern-density configuration.

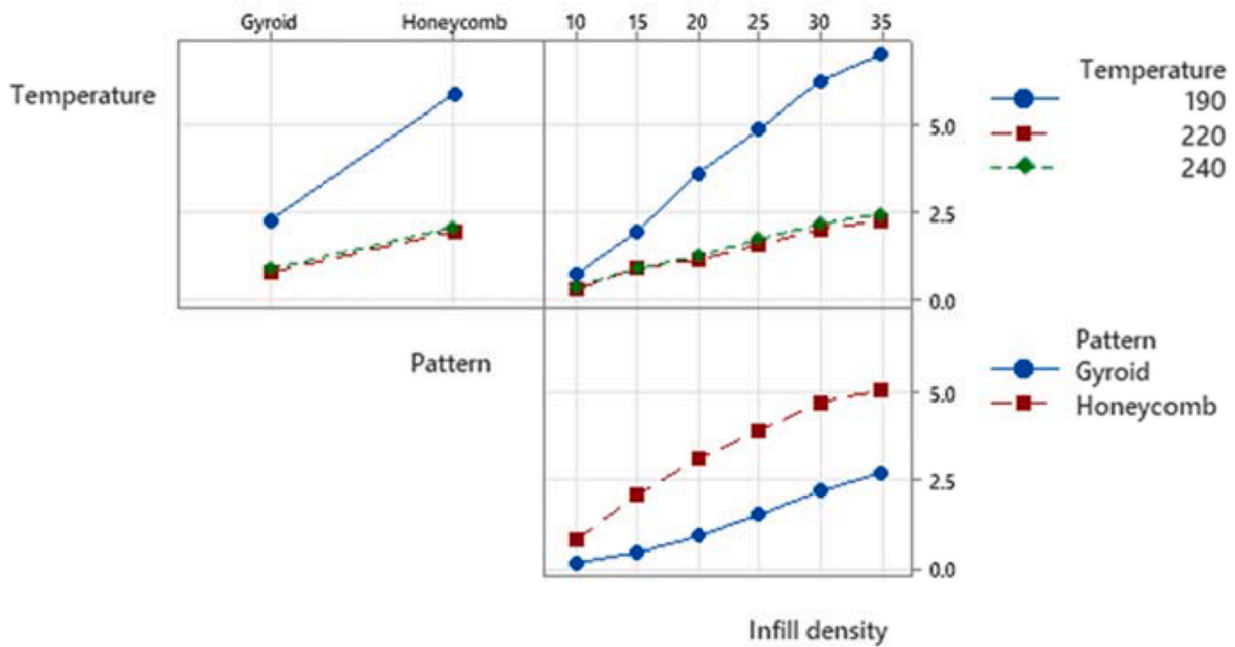
The plantar pressure values were higher when V3 was used as this insole included two zones with gyroid infill (G25, G15) – the gyroid pattern was less stiff than honeycomb, and one with honeycomb with low infill density (H10). This explains the peak pressure values of over 90 kPa, in comparison with V1 or V3 insoles.

V3 insole recorded the higher pressure around the toe (on one sensor) and heel (on two sensors), while for the other insoles the pressure was distributed also on the other sensors from the forefoot zone. Interesting conclusions can be drawn when comparing V3 with V4 insole (which was G20 uniform). In this case, we considered that the uniform gyroid infill allowed a better pressure distribution, this pattern being known as having an almost isotropic behavior [35].

The user feedback on comfort is detailed further. In comparison, V1 and V2 insoles were felt similarly in terms of pressure distribution. However, V2 was perceived as softer due to a larger area of gyroid pattern infill. In the areas with high plantar pressure, V2 felt firm due to the vertical walls which characterize the honeycomb pattern. On the contrary, V3 was notably softer, and that allowed the feet to comfortably settle on the insoles. The peak pressure zones addressed with gyroid infill were felt softer, which contributed to enhanced comfort, even if the plantar pressure was not fully reduced (approximately 35 % from the initial plantar pressure). V4 and V5 were quite distinct: V4 was comfortable and soft, creating the feeling that the insole molds to the sole of the foot. In contrast, V5 was stiffer, conveying the sensation that the infill undergoes minimal deformation under the influence of the user's weight. V3 insole received the most favorable feedback for achieving optimal comfort through pressure reduction. It balanced comfort and pressure reduction. The insole with gyroid pattern for high- and low-pressure areas effectively supports weight distribution and provides user a comfortable standing position.



a.



b.

Fig. 12. Interaction plots for 20 % strain (a) and 10 % strain (b) for compressive modulus.

4. Conclusions and further research

The customization of insoles and their stiffness are important aspects from both mechanical and cushioning perspectives, particularly for individuals with conditions such as diabetes, flexible flat foot, rigid foot, and plantar fasciitis, as well as for those who endure prolonged periods of standing.

Customizing stiffness according to varying plantar pressures in different foot zones improves the functionality and effectiveness of insoles. This study focused on examining a filament containing foaming agents, particularly varioShore TPU, for use in MEX process, as it offers

additional customization through adjustments in printing temperature. This complements the stiffness variations achieved by manipulating infill pattern and density settings. Although this type of filament offers specific customization properties in terms of stiffness, there are not many studies focused on better understanding and using its capabilities for comfort applications.

Mechanical tests assessed the impact of printing temperature, infill density, and infill pattern on compression behavior. Results identified printing temperature as the most influential factor, this finding being also demonstrated using SEM investigations. Correlations were established between peak plantar pressure values and infill pattern-density

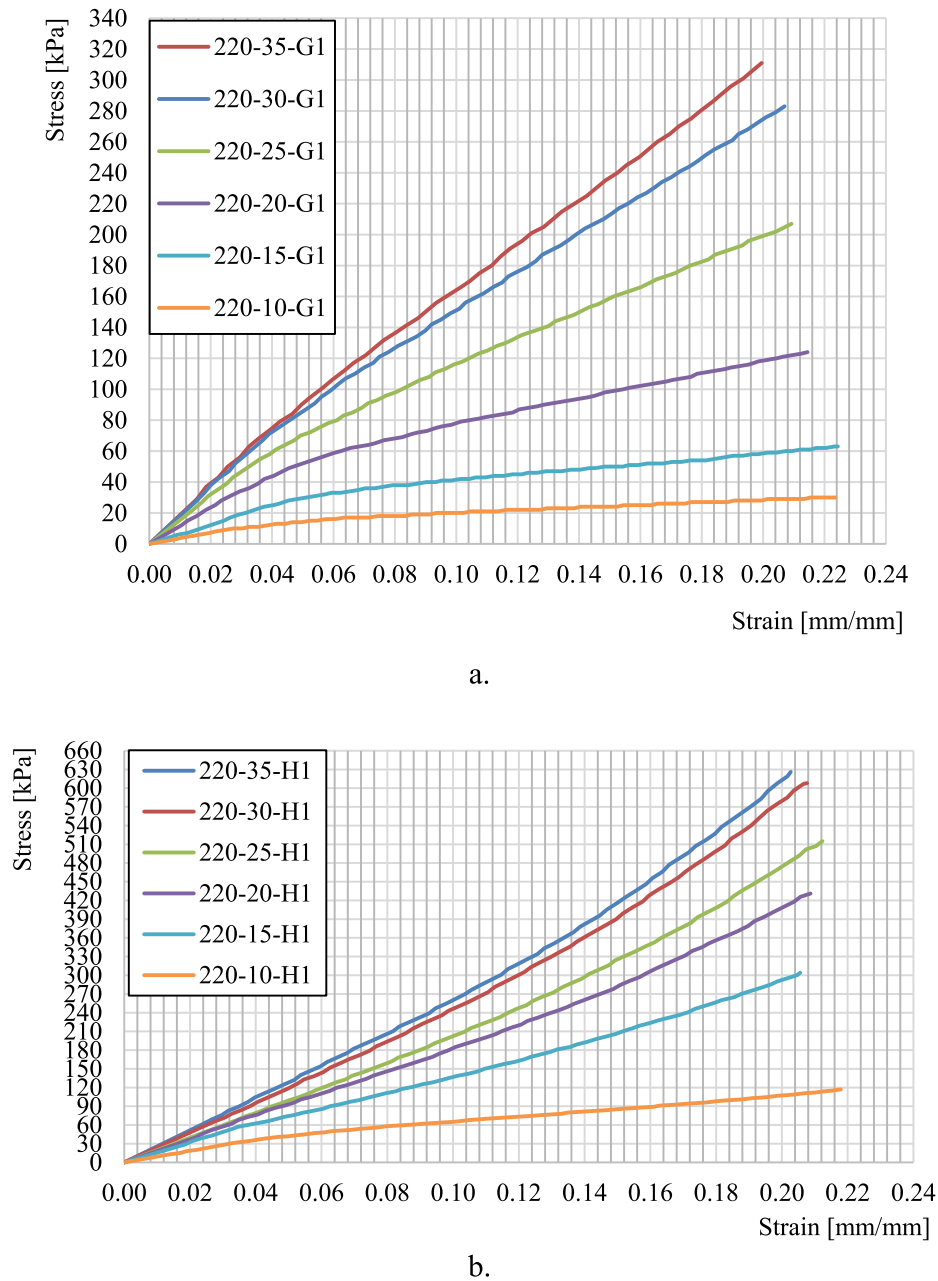


Fig. 13. Stress-strain curves for the last loading cycle: a) gyroid configuration; b) honeycomb configuration.

configurations. This data guided the customization of insoles for a healthy user, tested for comfort and plantar pressure distribution. Measurements confirmed the effectiveness of the developed methodology in achieving personalized and efficient foot orthotic solutions.

In this study, the decision to use 220 °C as printing temperature for the final insole design was based on several considerations. Firstly, this temperature was found in prior work to offer the proper balance between optimal foaming for comfort, and durability – and these were important aspects for ensuring user satisfaction. Secondly, prior work indicated that the insoles 3D-printed at 220 °C offered durability withstanding over 300,000 flexion cycles without any structural failure or deformation.

While the study primarily investigated on the static comfort of insoles printed at this temperature, it was acknowledged that varying printing temperatures allow further customization of the mechanical properties of the insoles. For instance, lower printing temperatures such as 190 °C (which produces stiffer prints), potentially benefiting those

individuals requiring more support due to a higher body weight. Thus, the stiffness in different zones of the same insole could be further tuned by using a 3D printer with two nozzles, which allows printing the material at different temperatures (e.g., one nozzle deposits filament at 220 °C in the heel area, while the other nozzle deposits filament at 190 °C in the midfoot zone), thereby achieving zones with custom compression strengths.

The results of compression testing at three different temperatures, infill densities and patterns provided information on the value of stiffness corresponding to these parameters combinations. This data can be used when developing a finite element model able predict the effects of stiffness variations due to parameter settings. The plantar pressure data gathered from conducting testing during standing, walking and running will be correlated with the values from stress-strain curves and compressive moduli to ensure the optimal uniform pressure distribution, enabling a more tailored approach to meet specific user needs.

Further research will be focused on gathering more experimental

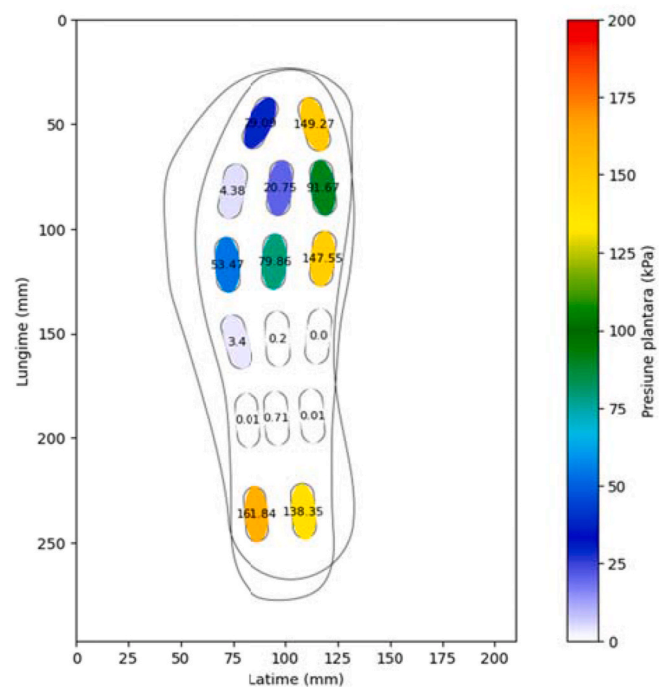


Fig. 14. Values of peak plantar pressure with user's bare foot.

Pressure value [kPa]	Pattern and infill density				
	Insole V1	Insole V2	Insole V3	Insole V4	Insole V5
161.84	H30	H25	G25	G20	H20
91.67	G25	H15	H10		
53.47	H10	G20	G15		

data for building a comprehensive finite element model of the insole, which in association with the peak plantar pressure data to allow a semi-automated customization of the infill variability. Establishing pressure distribution, dynamic peak pressure values, maximum admissible peak plantar pressure, as well as displacements are knowledge provided by the podiatrist based on patient's characteristics and pathology. Additionally, other expandable filaments will be studied, and their suitability for producing insoles will be evaluated by hardness studies, comfort evaluation, fatigue resistance or breathability. The aim is to offer enough engineering data for orthotics specialists to decide which filament better suits an individual in need of an insole with tailored stiffness.

CRediT authorship contribution statement

Mariana Cristiana Iacob: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Diana Popescu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis. **Contantin Stochioiu:** Writing – review & editing, Validation,

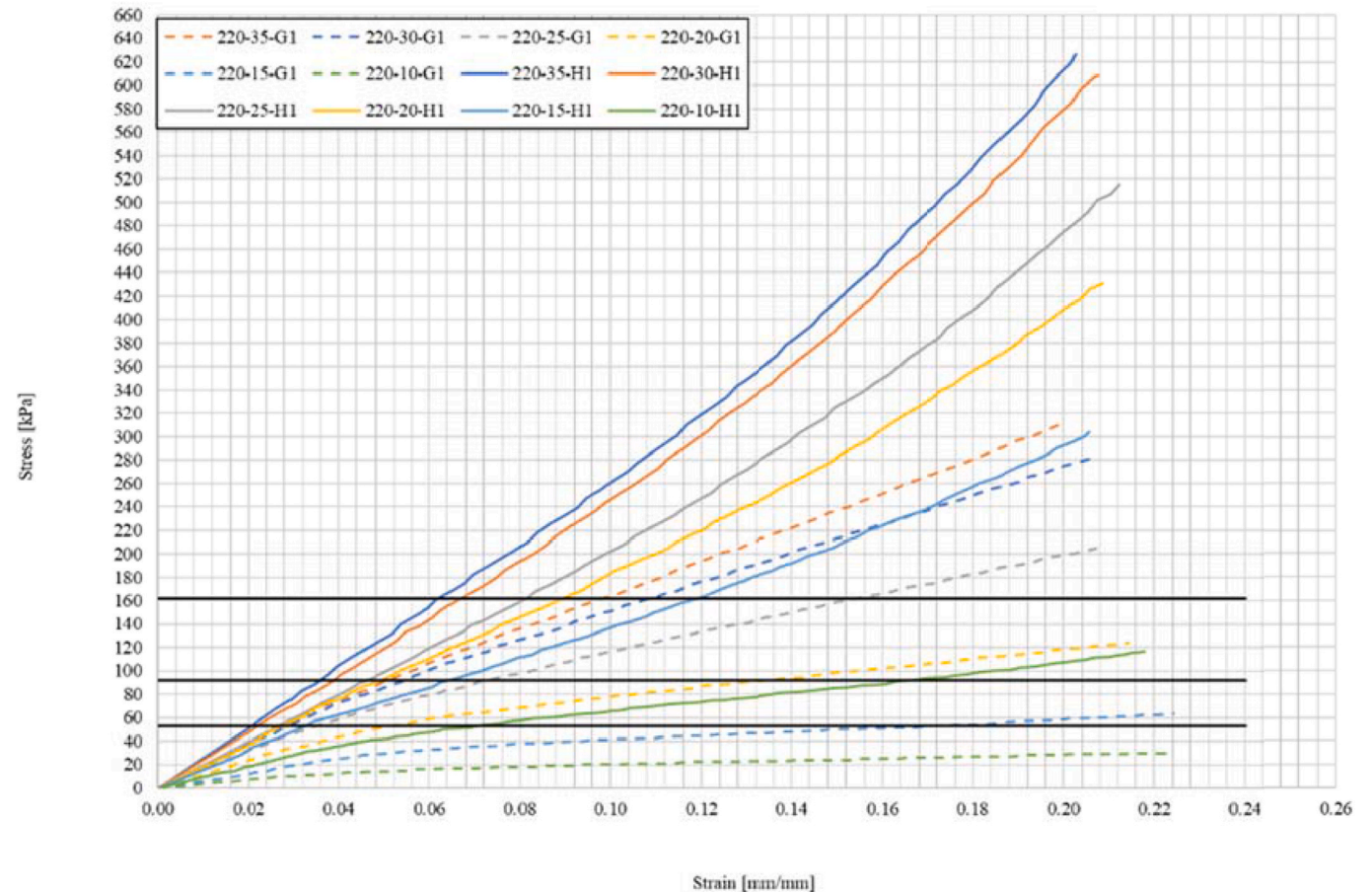


Fig. 15. Selecting the configuration pattern-infill density for different plantar pressures (the horizontal lines indicate the plantar pressure of interest).

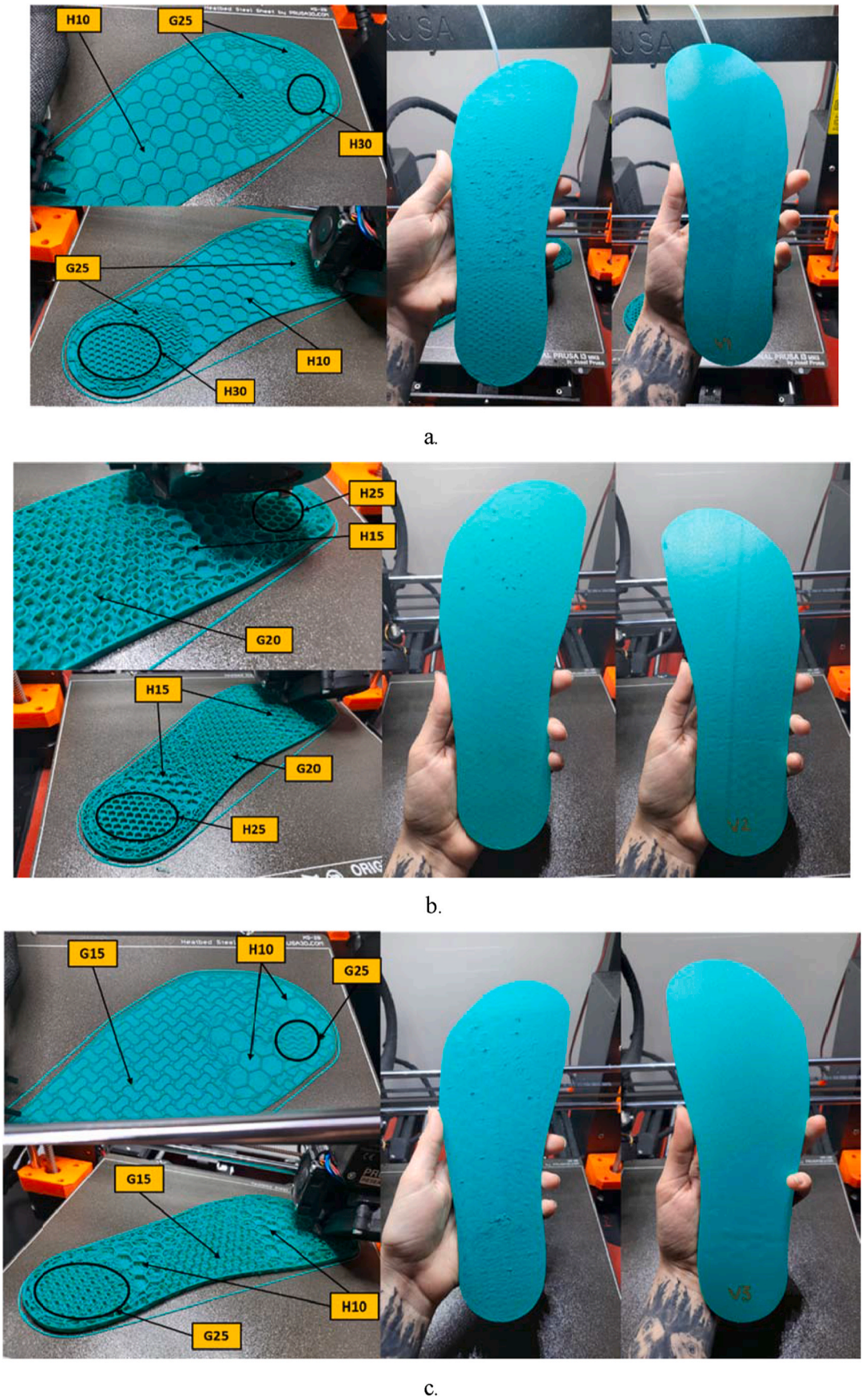


Fig. 16. Three of the 3D-printed insoles: a) version V1; b) version V2; c) version V3.

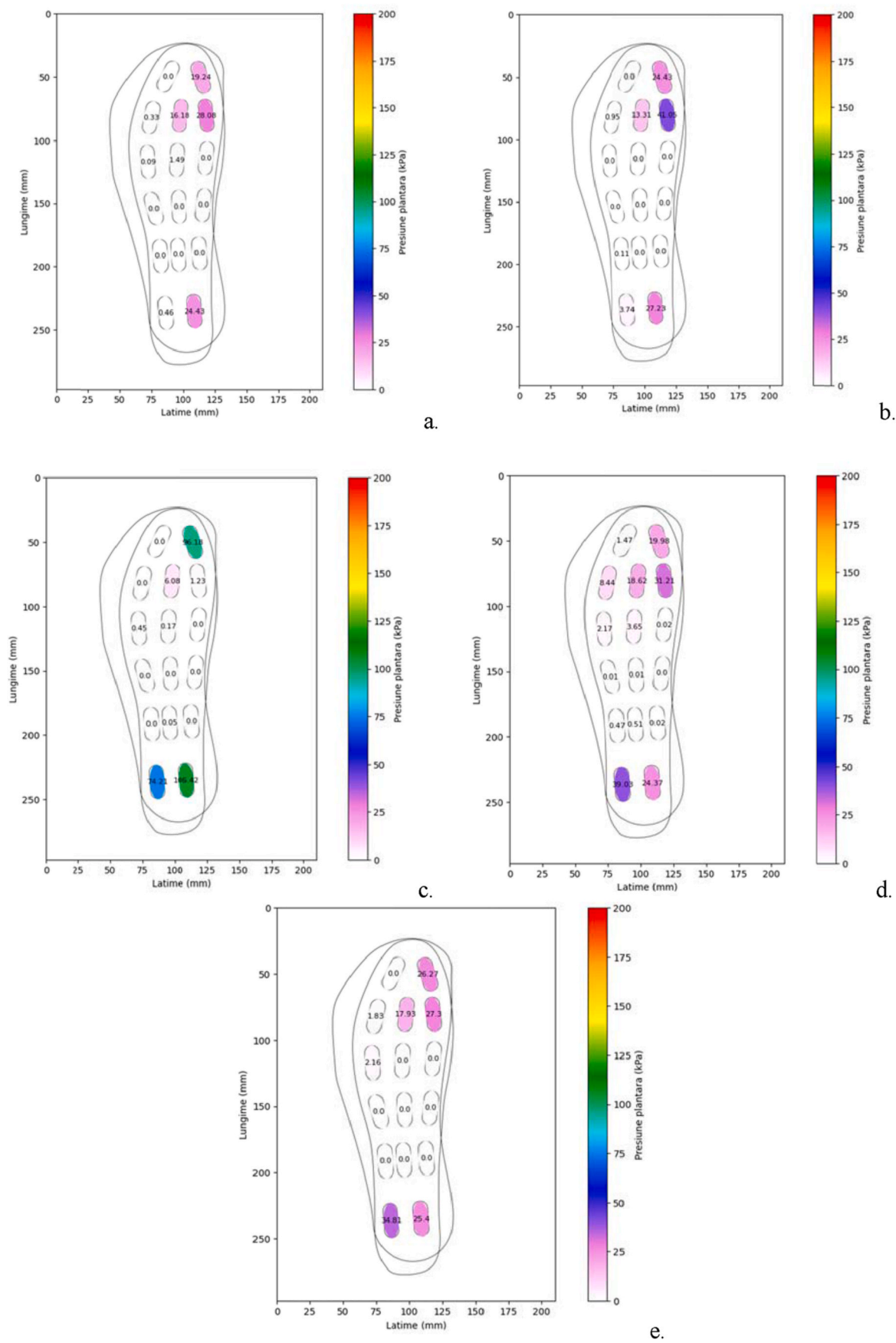


Fig. 17. Static plantar pressure results when using: a) V1 insole; b) V2 insole; c) V3 insole d) V4 insole; e) V5 insole.

Table 6

Peak pressure in heel and forefoot zones for each insole.

Insole	Heel peak pressure [kPa]	Forefoot peak pressure [kPa]
Bare foot	161.84	149.27
V1	19.24	24.43
V2	24.43	27.23
V3	96.18	106.42
V4	19.98	39.03
V5	26.27	34.81

Methodology, Investigation, Data curation. **Florin Baciuc**: Writing – original draft, Validation, Methodology, Investigation. **Anton Hadar**: Writing – review & editing, Validation, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data was added as supplementary material.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymertesting.2024.108517>.

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